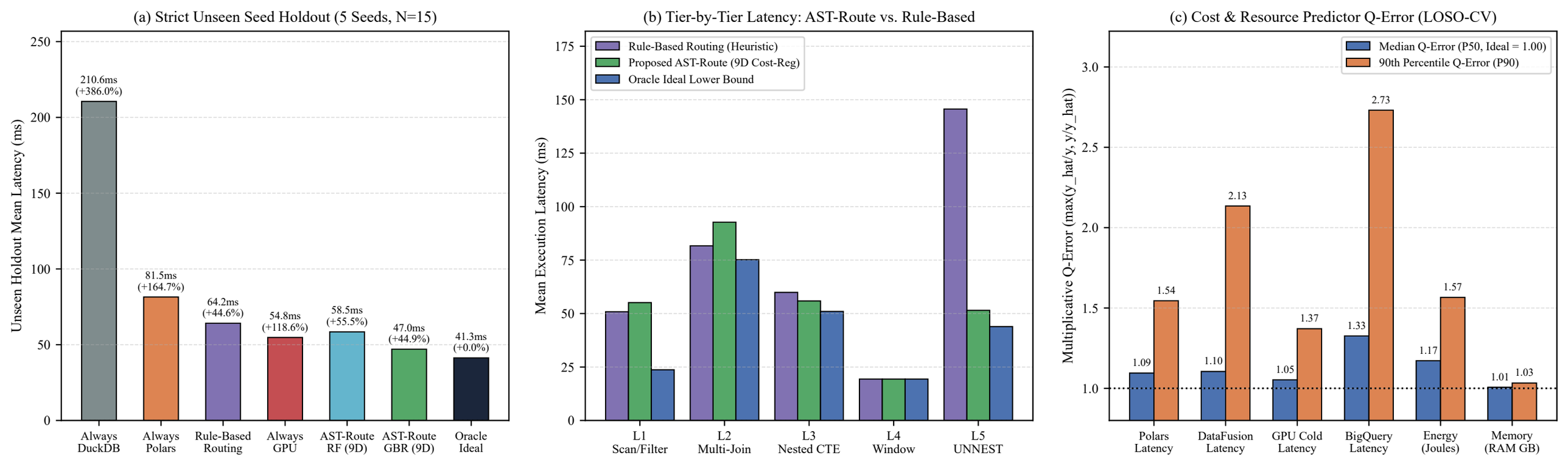
(a) Strict Unseen Seed Holdout (5 Seeds, N=15)
Unseen Holdout Mean Latency (ms)
0
50
100
150
200
250
210.6ms
(+386.0%)
81.5ms
(+164.7%)
64.2ms
(+44.6%)
54.8ms
(+118.6%)
58.5ms
(+55.5%)
47.0ms
(+44.9%)
41.3ms
(+0.0%)
Always DuckDB
Always Polars
Rule-Based Routing
Always GPU
AST-Route RF (9D)
AST-Route GBR (9D)
Oracle Ideal
(b) Tier-by-Tier Latency: AST-Route vs. Rule-Based
Mean Execution Latency (ms)
0
25
50
75
100
125
150
175
Rule-Based Routing (Heuristic)
Proposed AST-Route (9D Cost-Reg)
Oracle Ideal Lower Bound
L1 Scan/Filter
L2 Multi-Join
L3 Nested CTE
L4 Window
L5 UNNEST
(c) Cost & Resource Predictor Q-Error (LOSO-CV)
Multiplicative Q-Error (max(y_hat/y, y/y_hat))
1.0
1.5
2.0
2.5
3.0
Median Q-Error (P50, Ideal = 1.00)
90th Percentile Q-Error (P90)
1.09
1.54
1.10
2.13
1.05
1.37
1.33
2.73
1.17
1.57
1.01
1.03
Polars Latency
DataFusion Latency
GPU Cold Latency
BigQuery Latency
Energy (Joules)
Memory (RAM GB)